香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

2024-2025 Term 2

Final Examination

May 14th, 2025; Time Allowed: 3 Hours

NAME (print) _____

CUHKSZ ID _____

Room No. _____

Seat No. _____

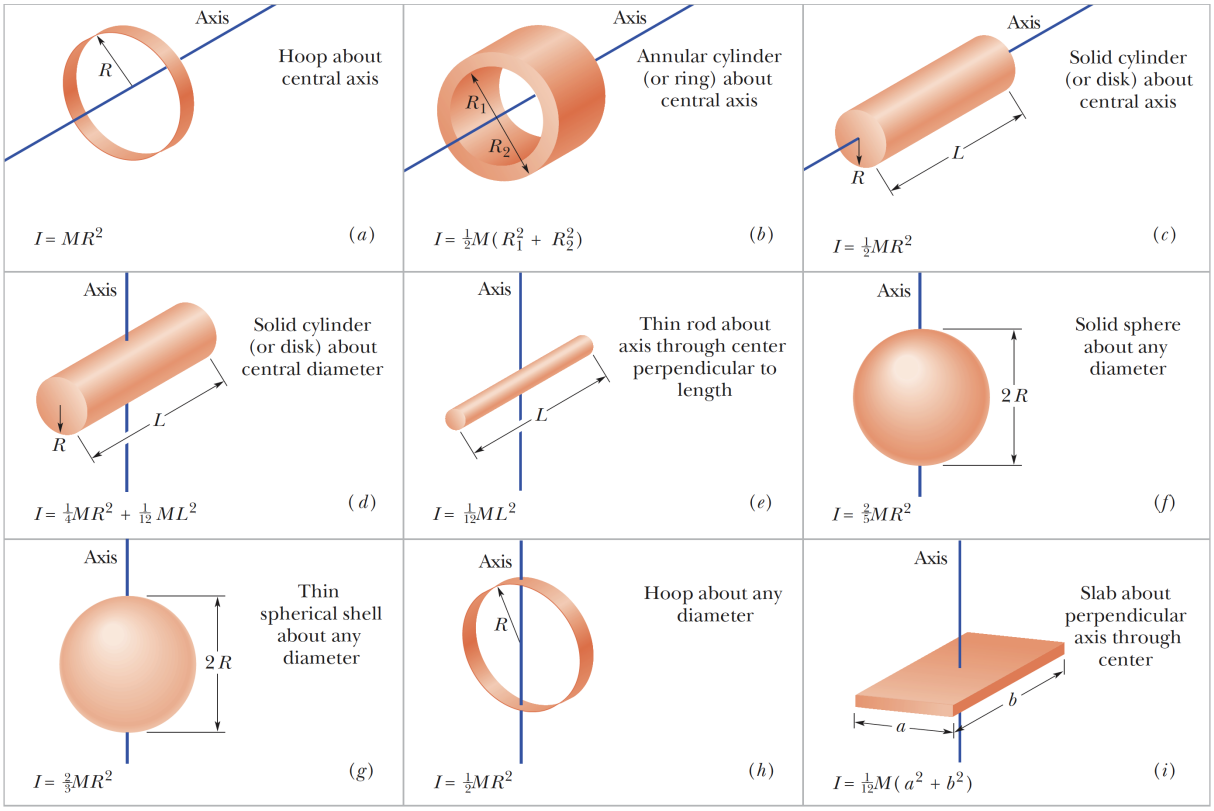
- **Show all your work**. Correct answers with little supporting work will not be given credit.
- Closed Book Exam: One piece of double-sided A4 reference paper, a scientific calculator, and a paper-based dictionary are allowed.
- Unless approved by the instructors, students who arrive more than 30 minutes late will NOT be admitted.
- The total points are 120 points. You need to finish ALL the questions in 3 hours (180 minutes).

#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	TOTAL
10	10	10	10	10	10	10	10	10	10	10	10	120

Summary of Basic Calculus:

$$\frac{d}{dx}x^n = nx^{n-1},$$
$$\frac{d}{dx}e^{ax} = ae^{ax},$$
$$\frac{d}{dx}\cos ax = -a\sin ax, \quad \frac{d}{dx}\sin ax = a\cos ax,$$
$$\frac{d}{dx}\ln ax = \frac{1}{x},$$
$$\frac{d}{dx}(uv) = v\frac{d}{dx}u + u\frac{d}{dx}v,$$
$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \ (n \neq -1),$$
$$\int \frac{dx}{x} = \ln x + C,$$
$$\int e^{ax}dx = \frac{1}{a}e^{ax} + C, \quad C \text{ is a constant.}$$

Table of Moments of Inertia:



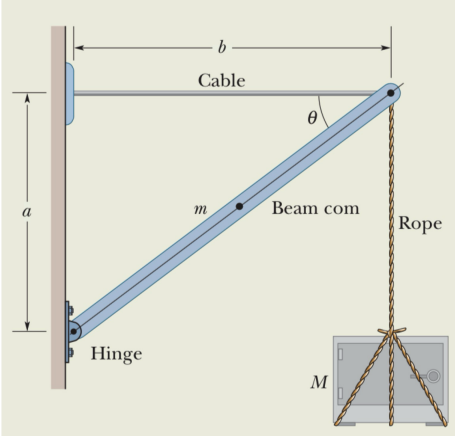
Bernoulli's Equation

$$P + \rho g y + \frac{1}{2}\rho v^2 = \text{constant.}$$

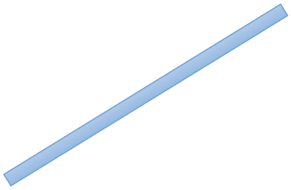
Spring Mass System: Simple Harmonic Oscillator

$$m\frac{d^2x}{dt^2} = -kx.$$

1. A block, of mass $M = 4.5 \times 10^2$ kg, is hanging by a rope from a boom with dimensions $a = 2.0$ m and $b = 2.5$ m. The boom consists of a hinged beam and a horizontal cable that connects the beam to a wall. The uniform beam has a mass $m = 80$ kg; the mass of the cable and rope are negligible. **(10 pts)**



(a) Draw the forces acting on the beam. **(2 pts)**

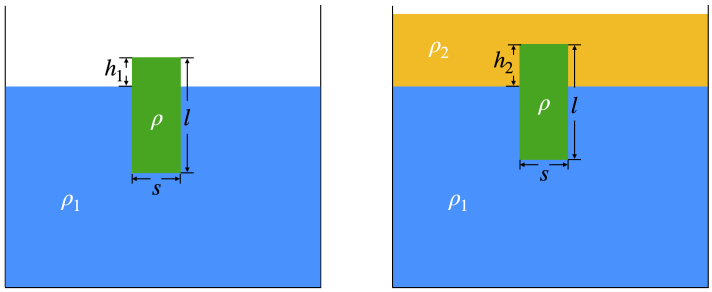


(b) What is the magnitude of tension T_c in the cable? **(2 pts)**

(c) Find the magnitude of net force on the beam from the hinge. **(4 pts)**

(d) Does the net force calculated in (c) point along the beam? Explain Why. **(2 pts)**

2. A cylinder of density ρ floats in equilibrium in a liquid of density ρ_1, as shown in the left figure below. The cylinder has a cross section s and length l. Suppose $\rho < \rho_1$. The gravitational acceleration constant is g, and the buoyancy of air can be ignored. **(10 pts)**



(a) Find the magnitude of the cylinder's weight. **(1 pt)**

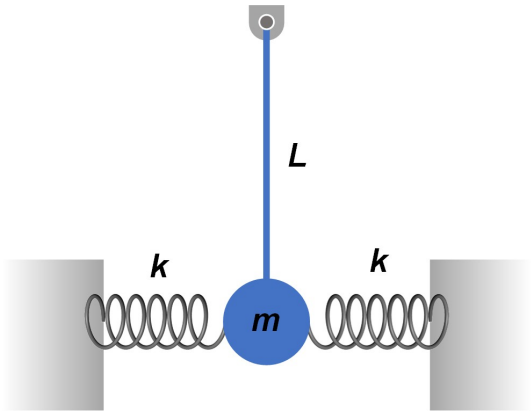
(b) Based on the fact that the cylinder is in equilibrium, find the buoyancy of liquid on the cylinder. Please specify magnitude and direction. **(2 pts)**

(c) The height of the cylinder above the liquid is h_1. According to Archimedes' principle, find the magnitude of buoyancy of liquid. **(1 pt)**

(d) Find h_1 in terms of the rest of the parameters. **(2 pts)**

(e) A second liquid of density ρ_2 is added to the container and immerses the cylinder, as shown in the right figure above. Suppose $\rho_2 < \rho$, and the cylinder is again in equilibrium. The cylinder's length in the second liquid is h_2. Find h_2. **(4 pts)**
Hint: You should find the same answer as (d) if you set $\rho_2 = 0$. In this way, you can double check your result.

3. For a pendulum coupled to two springs, as shown in the figure below, a bob of mass m is connected to a thin massless rod of length L and two weightless springs that have a force constant k. When the bob is directly below the pendulum support, the two springs are unstressed.

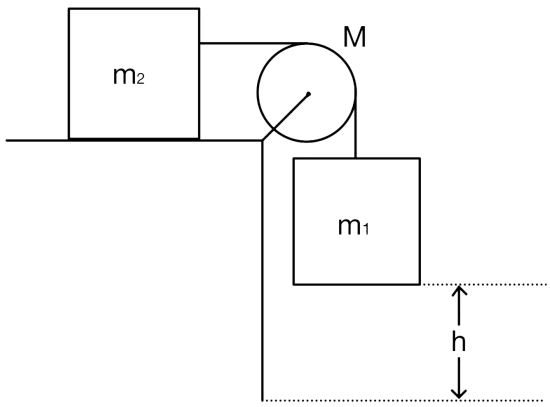


(a) Assuming the bob is slightly displaced from the equilibrium position and released for a small-amplitude vibration, write the equation of motion for the bob. **(4 pts)**

(b) Show that the oscillation is a simple harmonic motion. **(3 pts)**
Hint: Use the small-angle approximation.

(c) Find the period of this oscillating system for small-amplitude vibrations. **(3 pts)**

4. Two blocks, m_1 and m_2, are connected by a massless and non-stretchable cable that passes over a pulley. Block m_1 is released from rest at a height h above the floor; while block m_1 is placed on a frictionless desk. The pulley is a thin disk with a mass of M and a radius of R. It is mounted on the desk and rotates with negligible friction about a fixed axis. As the block m_1 falls, the block m_2 slides along the desk, and the cable moves without stretching or slipping. **(10 pts)**



If the pulley is massless ($M = 0$):

(a) Calculate the acceleration of each block as they start moving. **(1 pt)**

Hint: Specify both the magnitude and direction.

(b) Determine the magnitude of the tension force in the cable. **(1 pt)**

If the pulley has mass ($M \neq 0$):

(c) Calculate the acceleration of each block as they start moving. **(2 pts)**

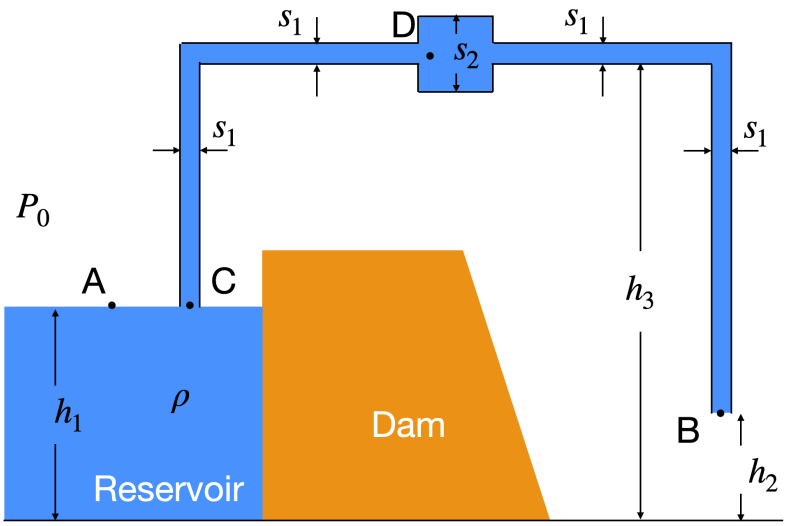
Hint: Specify both the magnitude and direction.

(d) Determine the magnitudes of tension forces in the cable on both sides of the pulley. **(2 pts)**

(e) When the falling block m_1 just reaches the ground, calculate the kinetic energies of the falling block m_1, the sliding block m_2, and the pulley M. **(3 pts)**

(f) What is the sum of all kinetic energy? Verify the plausibility of your answer. **(1 pt)**

5. A reservoir has water of depth h_1. An opening tube, known as siphon, is dipped into the water in the reservoir, and its other opening end is above the ground with height of h_2, as shown in the plot below. The cross section of the tube is s_1 everywhere, except an expanded region has across section s_2. The height of the horizontal part of the tube is h_3. The atmospheric pressure is P_0, water density is ρ, and gravitational acceleration constant is g. Suppose the reservoir is extremely large. **(10 pts)**



(a) Point A is at the reservoir surface and outside the tube. Find pressure P_A at point A. **(1 pt)**

(b) Point B is at the other opening end of the tube. Find pressure P_B at point B. **(1 pt)**

(c) Find water speed v_B at point B. **(2 pts)**

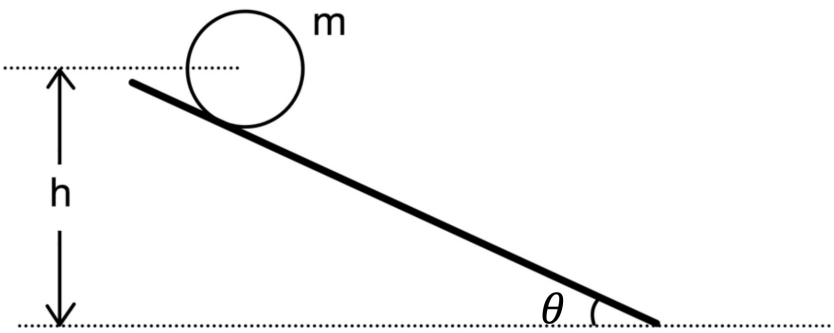
(d) Point C is at the reservoir surface and inside the tube. Find water speed v_C at point C. **(1 pt)**

(e) Find pressure P_C at point C. **(2 pts)**

(f) Point D is at the expanded region in the tube. Find water speed v_D at point D. **(1 pt)**

(g) Find pressure P_D at point D. Suppose cross section s_1 and s_2 are so small that any point in the horizontal part of the tube can be considered at the height h_3. **(2 pts)**

6. A uniform solid cylinder is rolling down a book that is tilted at an angle θ to the horizontal floor. The cylinder starts from rest at a vertical height h above the floor and rolls along the book. The book remains stationary throughout the motion. The cylinder has a mass m and a radius R. The coefficient of kinetic friction between the cylinder and the book is μ_k and the coefficient of static friction is μ_s. There exists a maximum critical angle θ_C at which the cylinder rolls without slipping. **(10 pts)**



(a) If $\theta < \theta_C$, the cylinder rolls smoothly down the book without slipping. Find the acceleration of the cylinder. **(1 pt)**

Hint: Specify both the magnitude and direction.

(b) Find the value of $\tan\theta_C$. **(2 pts)**

(c) If $\theta > \theta_C$, the cylinder begins to roll with slipping.

i. Is the friction involved static or kinetic friction? **(1 pt)**

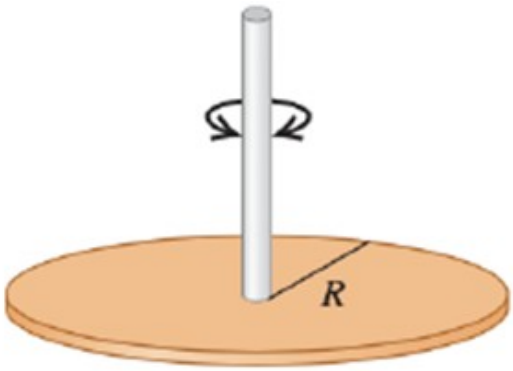
ii. Find the magnitude of acceleration of the cylinder's center of mass (COM) and its angular acceleration. Express the equation in terms of g, friction coefficients and angle θ. **(2 pts)**

iii. Find the distance L_{com} traveled by the cylinder's COM as it reaches the floor. **(1 pt)**
Hint: Distance is a scalar.

iv. Calculate the magnitude of angular displacement $\Delta\theta$ of the cylinder relative to its COM as it reaches the floor. **(1 pt)**

v. Verify whether $L_{com} = R\Delta\theta$ holds true or not in the above process. **(1 pt)**

7. A torsion pendulum consists of a disk with radius R and mass M, suspended by a thin wire with torsional constant κ. The disk is twisted by a small angle θ_m and released. The restoring torque is given by $\tau(t) = -\kappa\theta(t)$.



(a) Will this disk undergo simple harmonic motion after releasing? [Yes/No] **(1 pt)**

(b) Find the angular frequency Ω of the oscillation. **(2 pts)**

(c) Find the angular displacement $\theta(t)$ as a function of time. **(1 pt)**

(d) Find the angular velocity $\omega(t)$ of the disk as a function of time. **(1 pt)**

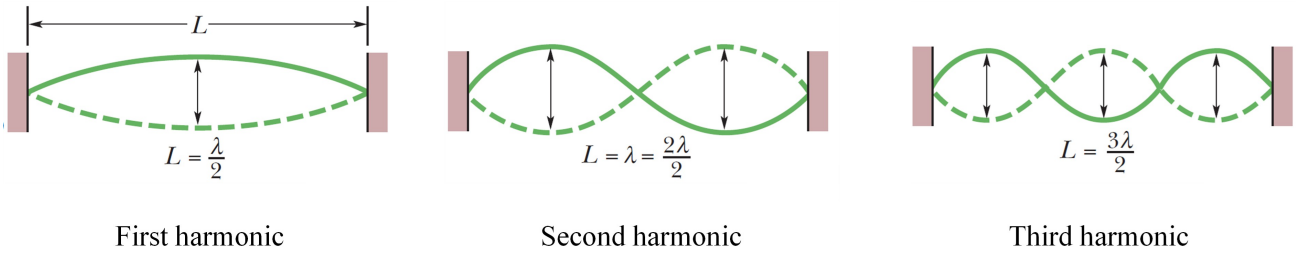
(e) Find the kinetic energy $K(t)$ of the disk as a function of time. **(1 pt)**

(f) Find the total mechanical energy of the oscillation. **(1 pt)**

(g) Assuming the motion is lightly damped with a damping torque $\tau_d = -b\omega$, write the modified equation of motion. **(2 pts)**

(h) An external oscillating torque $\tau_{ex} = Acos(\Omega_{ex}t)$ is applied to the torsion pendulum, leading to a driven oscillation. Assuming there is no damping, find the driving angular frequency that maximizes the vibration amplitude. **(1 pt)**

8. For a string fixed at both ends, the first three harmonics of standing waves are shown in the diagram below.



For the following types of pipes with length of L, fine the resonant conditions for the standing waves. The wave velocity is v_s.

(a) For a pipe with two open ends, draw the first three harmonics of standing sound waves, indicate the nodes and antinodes in each diagram. **(3 pts)**

(b) For a pipe with two open ends, find the resonant wavelength λ and frequency f for the Nth harmonic standing wave. **(2 pts)**

(c) For a pipe with one end closed, draw the first three harmonics of standing sound waves, indicate the nodes and antinodes in each diagram. **(3 pts)**

(d) For a pipe with one end closed, find the resonant wavelength λ and frequency f for the Nth harmonic standing wave. **(2 pts)**

9. The speed of longitudinal sound wave depends on the medium through which it travels and is described by the equation:

$$v = \sqrt{B/\rho}$$

where B is the bulk modulus, defined as $B = -\dfrac{\Delta p}{\Delta V/V}$.

However, does the temperature affect speed of sound? This question sparked a debate between two renowned physicists, Isaac Newton and Pierre-Simon Laplace. Let's now explore which theory aligns with reality.

Assume the sound wave travels in an ideal gas with a density of ρ at temperature T. The ideal gas law is given by:

$$pV = RT$$

where p is the pressure, V is the molar volume, and R is the ideal gas constant. **(10 pts)**

(a) Newton assumed that sound propagation was an isothermal (constant temperature) process. If the ideal gas is compressed while the temperature T is held constant, find the bulk modulus B. **(3 pts)**

(b) Laplace pointed out that the sound propagation is an adiabatic process (no heat exchange occurs during the process). When an ideal gas is compressed in adiabatic process, the pressure and volume are related by:

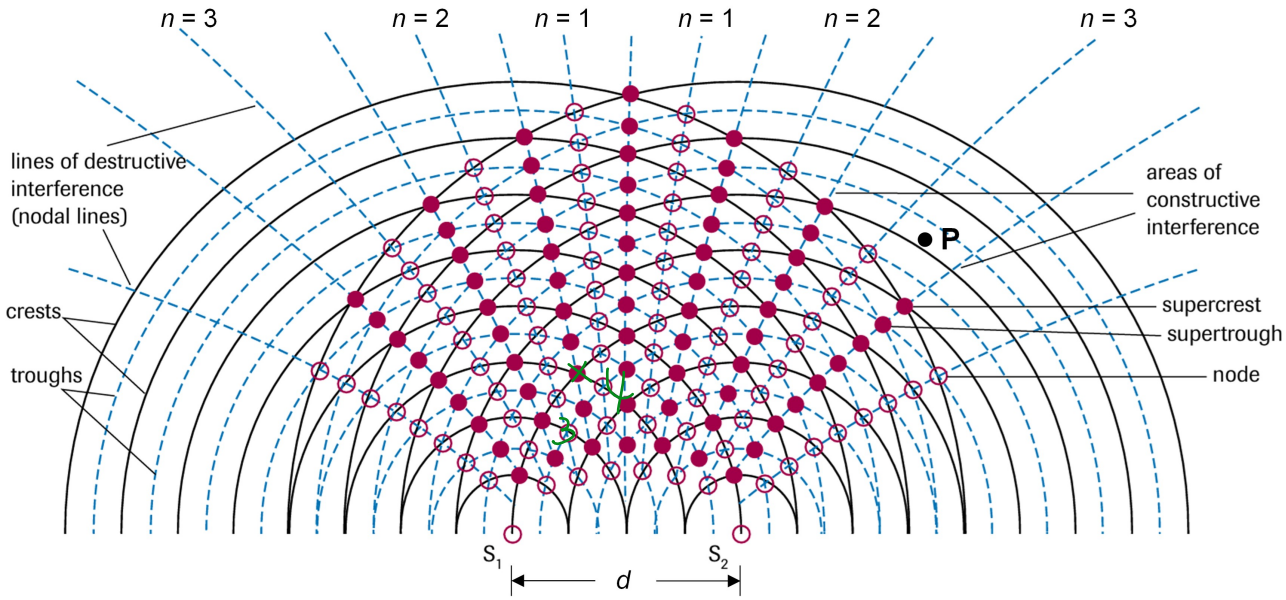
$$pV^{\gamma} = \text{constant}$$

where γ is the adiabatic constant having different values for different gases. Find the bulk modulus B in this adiabatic process. **(4 pts)**

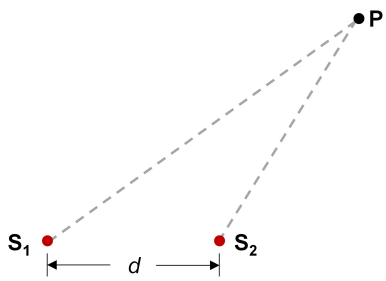
(c) At T = 300 K, the measured speed of sound in the air is approximately 346 m/s. The following parameters for dry air are provided: the density of air $\rho_{air} = 1.18\,kg/m^3$, the atmospheric pressure $p_0 = 1.01 \times 10^5$ Pa, and the adiabatic constant γ = 1.40. Calculate the speeds of sound according to Newton's isothermal theory and Laplace's adiabatic theory, respectively. **(2 pts)**

(d) Which speed is closer to the measurement? Justify this with physics explanation. **(1 pt)**

10. Two point sources, S_1 and S_2, are floating and oscillating on a calm lake and emitting surface waves with identical frequency f and wavelength λ. The distance between the sources is d. The waves propagate uniformly in all directions, creating a two-dimensional interference pattern. The solid circles represent antinodes, while the hollow circles denote nodes. The lines of destructive interference, labeled as $n = 1, 2, 3, \ldots$, are called nodal lines. Analyze the interference pattern and determine the conditions for constructive and destructive interference.



(a) Given the path difference $\Delta L = |PS_1 - PS_2|$, derive the condition for constructive interference at an arbitrary point P on the lake's surface when the two sources oscillate in phase, i.e., the two point sources have a phase difference of 0. (Hint: Express the condition in terms of ΔL and an integer n.) **(2 pts)**

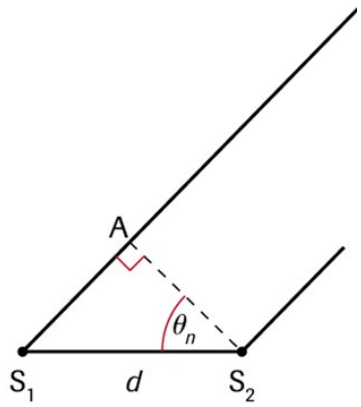


(b) Given the path difference ΔL, derive the condition for constructive interference at point P when the two sources oscillate out of phase, i.e., the two point sources have a phase difference of π. (Hint: Express the condition in terms of ΔL and an integer n.) **(2 pts)**

(c) Analyze the results above, does the interference pattern on the water's surface remain the same when the two sources oscillate in phase or out of phase? [Yes/No] **(1 pt)**

(d) Assuming the two sources oscillate in phase, when P is very far away compared to the separation distance d, the lines PS_1 and PS_2 are nearly parallel, and thereby we can define θ_n as the angle for the nth nodal line. Show that **(2 pts)**

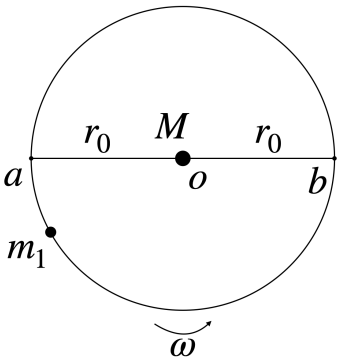
$$\sin\theta_n = (n - \frac{1}{2})\frac{\lambda}{d} \quad (n = 1, 2, 3, \ldots)$$

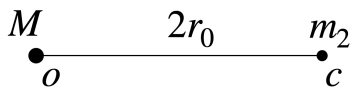


(e) Assuming the two sources oscillate in phase with $d = 3\lambda$, determine the number of nodal lines (lines of destructive interference) in the upper half-plane. **(2 pts)**
Hint: Consider the maximum and minimum path difference ΔL.

(f) Assuming the two sources oscillate in phase, determine the condition for the absence of nodal lines, i.e., the relationship between d and λ. **(1 pt)**

11. An object of mass m_1 circulates counterclockwise around a central object of mass M in a perfect circle. The circle has a radius of r_0. The central object M is located at point o, center of the circle. Point a and b are on the circle, and line ab is through point o, as shown in the left figure below. Gravitational constant is G. Suppose gravity is the only force considered in this question, $M \gg m_1$. **(10 pts)**





(a) Find the magnitude of angular velocity ω of object m_1. **(2 pts)**

(b) Find the time t_{ab} for object m_1 to travel from a to b, i.e. half a circle. **(1 pt)**

(c) In the right figure above, an object of mass M is located at point o. Point c is at a distance of $2r_0$ from point o. An object of mass m_2 is released from rest at point c and moves to M under gravity. Suppose $M \gg m_2$. When the object m_2 is at a distance of r from point o, write down the speed v of the object m_2 in terms of G, M, r, and r_0. **(2 pts)**

(d) Let us denote the time to travel from point c to point o is t_{co}. Define $u = \dfrac{r}{2r_0}$. Prove that

$$t_{co} = \sqrt{\frac{4r_0^3}{GM}} \int_0^1 \sqrt{\frac{u}{1-u}} du.$$ **(2 pts)**

(e) By changing variables from u to θ using $u = \sin^2\theta$, find t_{co}. **(2 pts)**

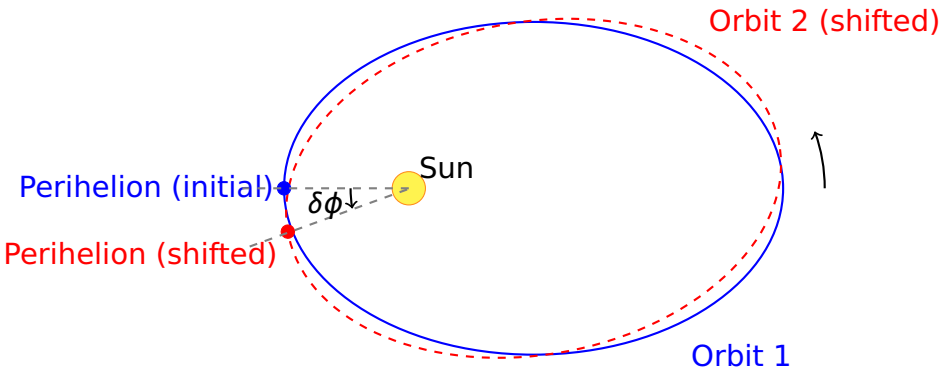
Hint: you may find the following trigonometric identities useful: $2\sin^2\theta = 1 - \cos 2\theta$.

(f) You may find that the answers from (b) and (e) are the same. Please justify this with a physics law. **(1 pt)**

Hint: You can view the motion from point c to point o as a motion from aphelion (farthest point) to perihelion (closest point) of an extremely flattened orbit. The orbit has a major axis (distance between aphelion and perihelion) of $2r_0$, minor axis of 0, and eccentricity $e = 1$.

12. Laplace-Runge-Lenz (LRL) Vector and Mercury's Orbital Precession **(10 pts)**

(DO NOT PANIC! No prior knowledge of the LRL vector or general relativity is required.)



The figure above is an illustration of the orbital precession of Mercury around the Sun. The size of $\delta\phi$ is exaggerated for illustrative purposes and does not represent the correct scale.

Historical Background of Mercury's Precession and Einstein's General Relativity:

- The precession of Mercury's orbit has been a cornerstone in modern physics, particularly in validating Albert Einstein's theory of General Relativity (GR).
- In the 19th century, astronomers observed that the perihelion (closest point to the Sun) of Mercury's orbit shifted over time, as illustrated in the figure above. Newtonian mechanics could not fully account for this phenomenon, known as anomalous perihelion precession.
- In 1915, Einstein proposed General Relativity, revolutionizing our understanding of gravity. Unlike Newtonian gravity, GR describes gravity as the curvature of spacetime caused by mass and energy. A key prediction of GR was the precise precession rate of planetary orbits – especially for objects like Mercury, which experiences a strong gravitational field near the Sun.
- The successful explanation of Mercury's precession became the first major triumph of General Relativity, marking a paradigm shift in physics.

This problem offers a glimpse into Einstein's monumental achievement. Traditionally, Mercury's precession is calculated by solving the equation of motion (Newton's second law) with relativistic corrections to the gravitational force. Here, we will employ an elegant slick approach using the Laplace-Runge-Lenz (LRL) vector, guiding you step-by-step through this fascinating phenomenon.

Laplace-Runge-Lenz (LRL) vector

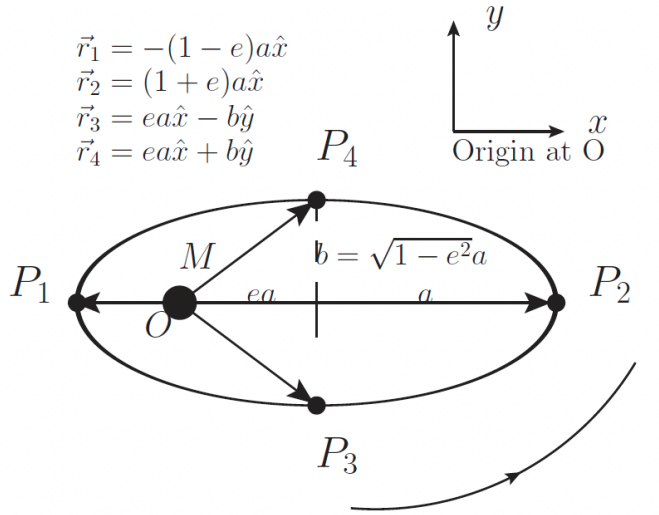
The LRL vector is an additional conserved quantity in Newtonian gravity for Keplerian orbits. For two celestial bodies interacting with Newton's gravitational force

$$\vec{F}_g = -\frac{GMm}{r^2}\frac{\vec{r}}{r} = -\frac{GMm}{r^2}\hat{r} \quad \text{with unit vector} \quad \hat{r} \equiv \frac{\vec{r}}{r},$$

the LRL vector remains constant regardless of where it is evaluated along the orbit. For the star-planet system shown below, the LRL vector is defined as

$$\vec{A} = \vec{v} \times \vec{L} - GMm\frac{\vec{r}}{r} = \vec{v} \times \vec{L} - GMm\hat{r},$$

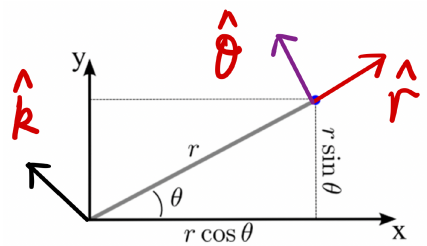
where $\vec{v}$ is the planet's orbital velocity and $\vec{L}$ is its angular momentum.



Consider the planet's elliptical orbit (shown above) with mass m rotating counterclockwise about a star of mass M. Assuming the star is sufficiently massive to remain approximately stationary at point O, the orbit has a semi-major axis a and eccentricity e.

Proof of Conservation of the LRL vector

For $\vec{F}_g = -\dfrac{GMm}{r^2}\hat{r}$, we demonstrate that the LRL vector $\vec{A}$ is conserved $\left(\dfrac{d\vec{A}}{dt} = 0\right)$ at all points on the orbit. In the polar (cylindrical) coordinate system shown below, the unit vectors satisfy the following relations:



$$\hat{r}\times\hat{\theta} = \hat{k}, \quad \hat{k}\times\hat{r} = \hat{\theta}, \quad \hat{\theta}\times\hat{k} = \hat{r};$$
$$\frac{d\hat{r}}{dt} = \omega\hat{\theta}, \quad \frac{d\hat{\theta}}{dt} = -\omega\hat{r}, \quad \frac{d\hat{k}}{dt} = 0.$$

One can use Newton's second law and conservation of $\vec{L}$ ($d\vec{L}/dt = 0$), and write as follows

$$\begin{aligned}\frac{d\vec{A}}{dt} &= \frac{d\left(\vec{v}\times\vec{L}\right)}{dt} - GMm\frac{d\hat{r}}{dt} = \frac{d\vec{v}}{dt}\times\vec{L} + \vec{v}\times\frac{d\vec{L}}{dt} - GMm\frac{d\hat{r}}{dt}\\ &= -\frac{GM}{r^2}\hat{r}\times(mr^2\omega\hat{k}) - GMm\omega\hat{\theta} = \frac{GMm\omega r^2}{r^2}\hat{\theta} - GMm\omega\hat{\theta} = 0,\end{aligned}$$

where we have expanded the first term and substituted $\vec{L} = \vec{r}\times\vec{p} = \vec{r}\times m(v_r\hat{r} + r\omega\hat{\theta}) = mr^2\omega\hat{k}$. For the second term, $\dfrac{d\hat{r}}{dt} = \omega\hat{\theta}$ (from polar coordinate kinematics).

(a) For Mercury's counterclockwise orbit around the Sun, <u>determine</u> the direction of the planet's angular momentum $\vec{L} \equiv \vec{r}\times\vec{p}$. **(1 pt)**

(b) Using the Sun as reference, prove that $\vec{L}$ is conserved ($\frac{d\vec{L}}{dt} = 0$) under the gravitational force along $\vec{r}$. **(1 pt)**

(c) Given Mercury's perihelion speed $v_p = \sqrt{\frac{GM}{a}\frac{1+e}{1-e}}$ and aphelion speed $v_a = \sqrt{\frac{GM}{a}\frac{1-e}{1+e}}$, calculate the LRL vector $\vec{A}$. **(1 pt)**

Hint: Since $\vec{A}$ is conserved, evaluate at the perihelion P_1 or the aphelion P_2 for simplicity. Your solution should confirm that $\vec{A}$ points along $-\hat{x}$.

(d) In Einstein's GR, the modified gravitational force law becomes

$$\vec{F}_{GR} = -\frac{GMm}{r^2}\left(1 + \frac{3L^2}{m^2c^2r^2}\right)\hat{r},$$

where $\frac{3L^2}{m^2c^2r^2} \ll 1$. While $\vec{L}$ remains conserved, $\vec{A}$ is not conserved now. For the Sun-Mercury system under $\vec{F}_{GR}$, $\frac{d\vec{A}}{dt} = D\frac{GML^2\omega}{mc^2r^2}\hat{\theta}$. Find the <u>coefficient D</u>. **(1 pt)**

Hint: Use $\vec{L} = mr^2\omega\hat{\theta}$ to calculate $\frac{d\vec{A}}{dt}$.

(e) Project $\frac{d\vec{A}}{dt}$ onto the x- and y-axes, then integrate both components over one orbital period to find δA_x and δA_y, defined by $\delta\vec{A} = \delta A_x \hat{x} + \delta A_y \hat{y}$. **(3 pts)**

Hint: Use $d\theta = \omega dt$, then integrate θ from 0 to 2π for one orbit. The orbit can be approximated by Newtonian solution with $\vec{A}$ along the $-\hat{x}$ direction:

$$r(\theta) = \frac{a(1-e^2)}{1-e\cos\theta},$$

as GR corrections to the orbit are negligible. You may also find the following integral identities useful:

$$\int_0^{2\pi} (1-e\cos\theta)^2 \cos\theta d\theta = -2\pi e \quad \text{and} \quad \int_0^{2\pi} (1-e\cos\theta)^2 \sin\theta d\theta = 0.$$

(f) Assume that $\vec{A}$ is along the $-\hat{x}$ direction, <u>show</u> that the perihelion precession angle per orbit, namely the angle that the LRL vector $\vec{A}$ rotates per orbit, is given by **(2 pts)**

$$\delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}.$$

(g) Given $1\,\text{rad} = 206265\,\text{arcsec}$ and:

$$\begin{aligned} G &= 6.67 \times 10^{-11}\,\text{m}^3\text{kg}^{-1}\text{s}^{-2} \\ M &= 2.0 \times 10^{30}\,\text{kg} \\ c &= 3.0 \times 10^{8}\,\text{m/s} \\ a &= 5.8 \times 10^{10}\,\text{m} \\ e &= 0.21 \\ T_m &= 0.24\,\text{years}, \end{aligned}$$

<u>compute</u> $\delta\phi$ in units of arcseconds/century. **(1 pt)**